

VULNERABILITY ASSESSMENT REPORT

VoteVerse Web Application Security Audit

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Target Website: <https://voteverse-2a625.web.app/>

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Executive Summary

This report presents the results of a web application vulnerability assessment conducted on the VoteVerse web application hosted on Firebase. The assessment focused on identifying browser-level security weaknesses, missing security headers, and configuration issues using OWASP ZAP and manual inspection techniques.

The assessment identified four primary security observations consisting of three medium-risk findings and one low-risk finding. No critical vulnerabilities such as SQL Injection, Remote Code Execution, or Cross-Site Scripting were identified during the assessment.

Severity	Count	Status
Medium	3	Open
Low	1	Open

Table of Contents

Executive Summary.....	2
1. Introduction	4
2. Objective	4
3. Tools Used.....	4
4. Scope of Assessment	4
5. Methodology.....	4
6. Findings Summary.....	5
7. Finding 1 — CSP Header Not Set.....	6
8. Finding 2 — Missing Anti-Clickjacking Header.....	7
9. Finding 3 — Sub Resource Integrity Missing.....	8
10. Finding 4 — X-Content-Type-Options Header Missing.....	9
11. Conclusion.....	10

1. Introduction

This report presents the results of a web application vulnerability assessment performed on the VoteVerse web application. The purpose of this assessment was to identify common security weaknesses, analyze their potential impact, and recommend remediation measures.

The assessment focused on browser security configurations, security headers, and passive vulnerability observations using OWASP ZAP.

2. Objective

- Identify common web application vulnerabilities
- Analyze security risks and security posture
- Evaluate browser security protections
- Recommend improvements for enhanced security

3. Tools Used

- OWASP ZAP (Zed Attack Proxy)
- Browser Developer Tools
- Firebase Hosting
- GitHub

4. Scope of Assessment

Target Website:

<https://voteverse-2a625.web.app/>

The assessment focused only on publicly accessible application components and passive vulnerability analysis.

5. Methodology

The following methodology was followed during the assessment:

- Reconnaissance and application accessibility testing
- Passive vulnerability scanning using OWASP ZAP
- Security header analysis
- Risk classification and validation
- Vulnerability documentation
- Remediation recommendation preparation

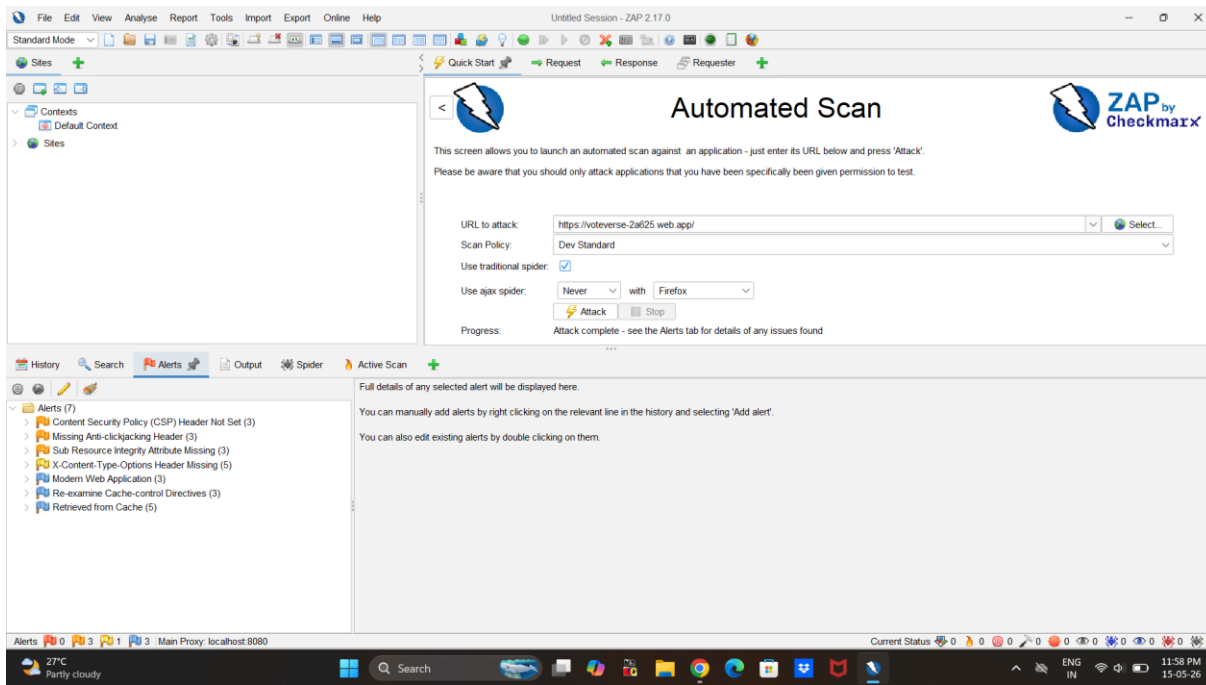


Figure 5: Overall OWASP ZAP vulnerability assessment scan results.

6. Findings Summary

Vulnerability	Risk Level	Status
Content Security Policy (CSP) Header Not Set	Medium	Open
Missing Anti-Clickjacking Header	Medium	Open
Sub Resource Integrity Attribute Missing	Medium	Open
X-Content-Type-Options Header Missing	Low	Open

7. Finding 1 — Content Security Policy (CSP) Header Not Set

Risk Level

Medium

Description

The application does not define a Content Security Policy (CSP) header. CSP helps browsers prevent malicious script execution and reduces the impact of Cross-Site Scripting (XSS) attacks.

Potential Impact

- Increased exposure to malicious script injection
- Reduced browser-level protection
- Increased risk of client-side attacks

Recommendation

Implement a strong Content Security Policy restricting allowed scripts, styles, and external resources.

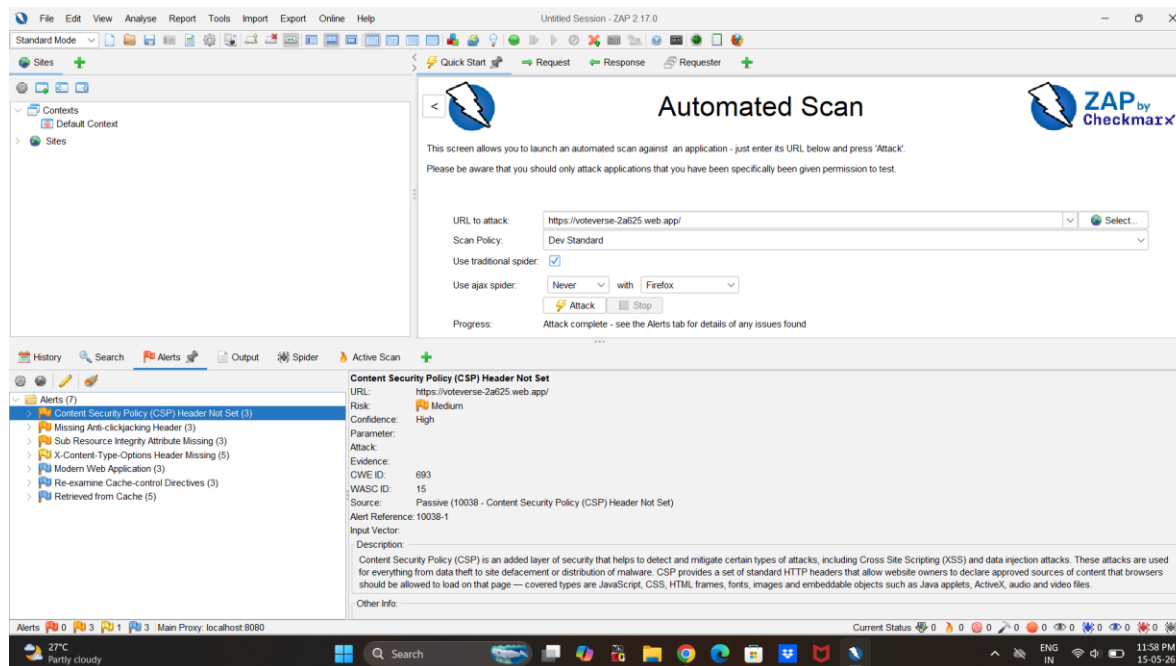


Figure 1: CSP Header vulnerability identified using OWASP ZAP.

8. Finding 2 — Missing Anti-Clickjacking Header

Risk Level

Medium

Description

The application does not implement protection against clickjacking attacks using X-Frame-Options or CSP frame-ancestors directives.

Potential Impact

- The website may be embedded inside malicious iframes
- Users may unknowingly interact with hidden elements

Recommendation

Implement X-Frame-Options: DENY or configure the CSP frame-ancestors directive.

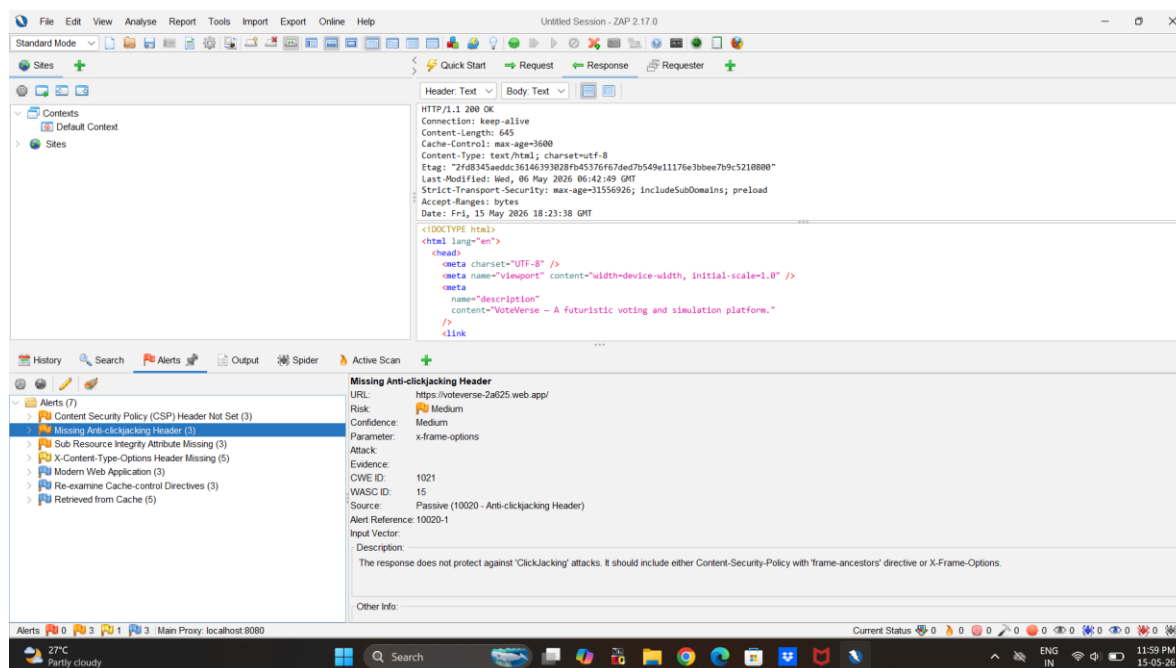


Figure 2: Missing Anti-Clickjacking Header identified during vulnerability assessment.

9. Finding 3 — Sub Resource Integrity Attribute Missing

Risk Level

Medium

Description

External resources such as scripts or stylesheets do not use Subresource Integrity (SRI) attributes.

Potential Impact

- If external CDN resources become compromised, malicious code may execute inside the application.

Recommendation

Add integrity and crossorigin attributes to externally loaded resources.

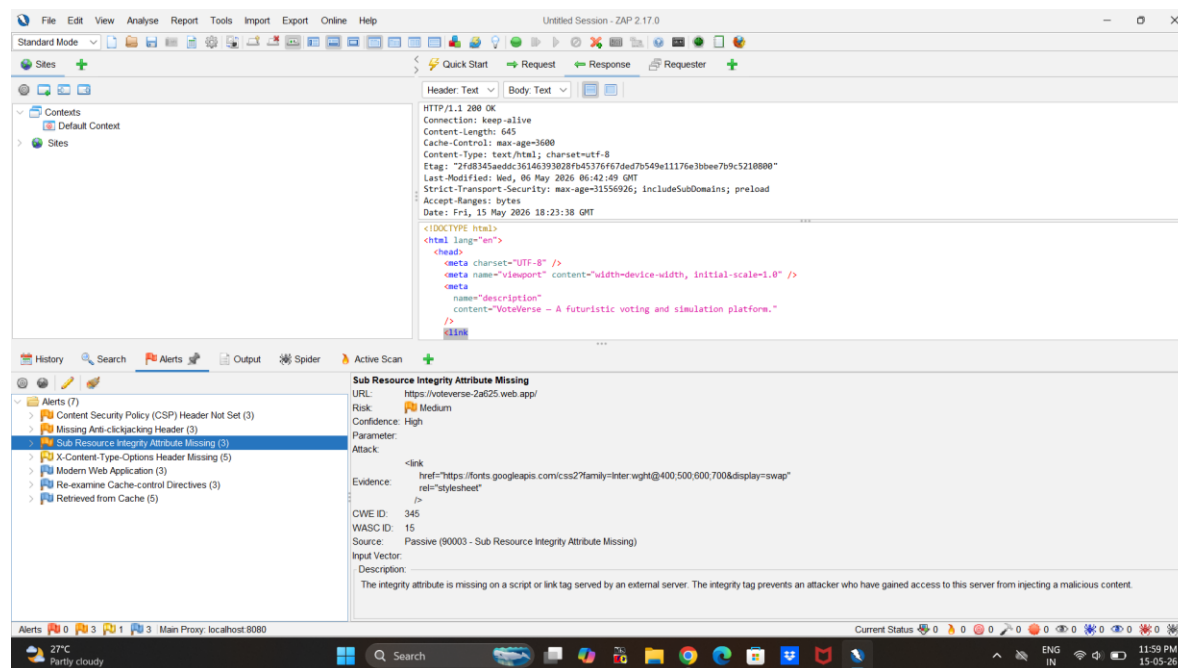


Figure 3: Missing Sub Resource Integrity protection identified in external resources.

10. Finding 4 — X-Content-Type-Options Header Missing

Risk Level

Low

Description

The application does not include the X-Content-Type-Options security header.

Potential Impact

- Browsers may incorrectly interpret content types, increasing MIME-sniffing related security risks.

Recommendation

Add the following response header: X-Content-Type-Options: nosniff

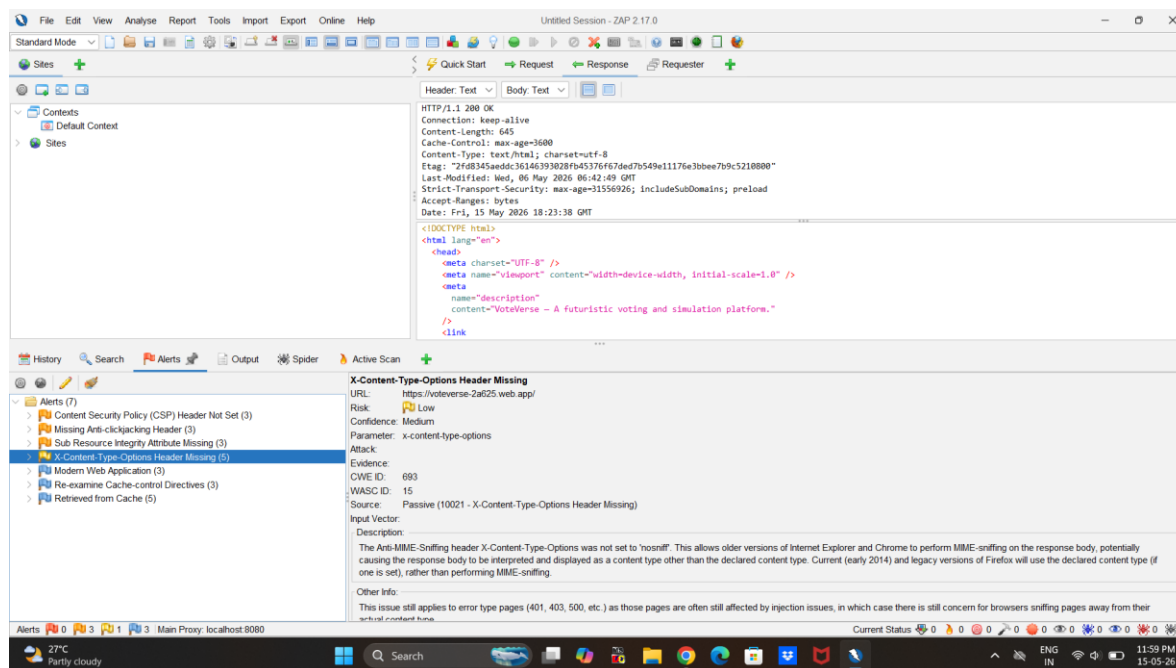


Figure 4: Missing X-Content-Type-Options security header identified using OWASP ZAP.

11. Conclusion

The vulnerability assessment identified several medium and low-risk browser security configuration issues within the VoteVerse web application.

No critical vulnerabilities such as SQL Injection or Cross-Site Scripting were identified during the assessment.

The application demonstrates a relatively secure foundation; however, implementing recommended browser security headers and integrity protections would further strengthen the overall security posture.

This assessment provided practical exposure to web application vulnerability analysis, risk assessment, and professional cybersecurity reporting.